

Εθνικό και Καποδιστριακό Πανεπιστήμιο Αθηνών
Τμήμα Πληροφορικής και Τηλεπικοινωνιών

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1 Data Preprocessing and Initial Event Replayer Implementation

The main data source utilized for this project is StatsBomb. StatsBomb is an open online database that provides detailed records of football matches in `.json` format. Our initial task was to filter the necessary information required for the implementation of the final system. The extracted fields are:

- milliseconds (from the start of the game)
- clock time (view of the clock)
- x coordinate of the ball
- y coordinate of the ball
- description of the event

To extract this information, a Python script (`event_reader.py`) was developed. Utilizing the `json` library, it successfully parses the necessary data and saves it into a text file. Additionally, to achieve smooth data playback, the halftime break was bypassed to maintain a continuous timeline.

Analyzing a sample of 5 different matches, it is observed that an average of approximately 3,562 events are recorded per match. The total number of events varies depending on the pace of each game, ranging from 3,205 to 3,910 events in our sample. As mentioned, these events do not have a constant recording rate; rather, they are distributed over time based on the actual actions (e.g., passes, shots) performed with the ball. This is why there is often a gap of a few seconds in the regular flow (e.g., the clock might jump from 37:10 to 37:12). This occurs because the available data is strictly event-driven, meaning time elapses without any new records until the next distinct action takes place.